Data: is the super power of this modern world , where we can solve many problems using data and can enhance our business or organization.

It is rightly said that data is the new currency of the modern world.

Data is the key ingredient to make any business or innovation excel, as data is the main ingredient for successful decision-making.

The better the quality of data, the higher the quality of the analysis, which results in effective and successful outcomes.

Data: It is a collection of raw information that does not have any meaning or context. Data can be present in the form of images , words, text numbers. Basically , Data are unprocessed facts and figures that do not hold any meaning.
Eg: {20,24, 25, 26, 27,21, 23 ,20, 24, 26}
Age of the students in the class : {20,24, 25, 26, 27,21, 23 ,20, 24, 26}

The youngest student in my class is of which age: 20 yrs
Eldest student in my class is of which age: 27 yrs
Avg age:

Information: Data would be considered as information when it is kept together in structured ways and once the data is in structured format it is called information . The information is a result of analyzing, organizing, and interpreting the data.

Eg: A company might use sales data to create a report that shows the avg sale, which product was the best seller, and which product got the lowest rating.

Knowledge: It is understanding of information and ability to see the relation between different pieces of information and draw conclusions.

Eg: A company might use their sales data to understand the factors that are driving their sales and to improve their marketing strategy .

Wisdom: This is the ability to apply knowledge to real-world scenarios in which we make decisions based on the knowledge and understanding that we have gathered up to this stage.

Eg: A company, after seeing the sales report, will draw the conclusion about which product they should launch or expand in which geographical regions.

Data is further divided into 2 types, i.e

1. **Structured Data**
2. **Unstructured data**
3. **Semistructured Data**

Structured Data:

These are the types of data that have a predefined model or schema to organize the data in tabular format with rows and columns.

Eg: Spreadsheet, Constant List, Relational Databases.

Unstructured data : These are those types of data that do not have any predefined format. It can include text, images, videos, emails, etc. This kind of data can have multiple varieties of formats and types.

Eg: Image, Audio, Video, Email

Semi-structured Data:

These are those types of data that are not in tabular format or row-column format but are organized properly so that we can easily use or analyze the data.

Eg: JSON, log files, XML files.

Can we store unstructured or semi-structured data in MySQL?

```
CREATE TABLE students (
   id INT PRIMARY KEY,
   name VARCHAR(100),
   profile JSON  -- semi-structured field
);

INSERT INTO students VALUES (
   1, 'Rahul',
   '{"skills": ["Python", "SQL"], "score": 92, "address": {"city": "Mumbai"}}'
);

-- Query inside JSON
SELECT name, profile->>'$.address.city' AS city FROM students;
```

```
-- Filter on JSON field
SELECT * FROM students WHERE profile->>'$.score' > 90;
```

File System vs DBMS – Drawbacks of File Systems

A **traditional file system** stores data in separate files managed by individual applications. As the amount of data grows, several problems occur.

Major Drawbacks of File Systems

No.	Drawback	Explanation
1	**Data Redundancy**	The same data may be stored in multiple files, resulting in unnecessary duplication.
2	**Data Inconsistency**	If data is updated in one file but not in other related files, different files may contain different values.
3	**Difficulty in Data Access**	Retrieving required information may require writing a new program for each type of query.
4	**Data Isolation**	Data is scattered across different files and formats, making it difficult to combine and access.
5	**Integrity Problems**	Maintaining rules such as salary > 0 or age >= 18 is difficult because constraints are usually handled by application programs.
6	**Security Problems**	File systems generally provide limited and less flexible control over who can access or modify specific data.

7	**Concurrent Access Problems**	When multiple users access or update the same file simultaneously, data may become inconsistent or corrupted.
8	**Atomicity Problems**	If a transaction fails halfway, it can be difficult to restore the data to its previous consistent state.
9	**Backup and Recovery Problems**	Recovering data after system failure can be difficult and may require manual procedures.
10	**Program-Data Dependence**	Changes in the file structure may require changes to the application programs that use those files.
11	**Poor Data Sharing**	Different applications may maintain separate files, making sharing of common data difficult.
12	**Limited Scalability**	Managing very large amounts of data becomes difficult and inefficient.

The major drawbacks of file systems are data redundancy, data inconsistency, data isolation, difficult data access, integrity and security problems, concurrent-access issues, atomicity problems, and poor backup/recovery.

There are the following types of file systems:

1. FAT: File Allocation Table
   These were used in **older Operating systems**

2. NTFS: New Technology File System:
   These are used in modern operating systems

3. HFS: Hierarchical File System
   These are used in **Mac OS**

4. **Ext4:**
   These are used in **Linux distributions**

ANSI definition of DBMS -> collection of programs that allows you to insert, update, delete, and process

Difference between File System and DBMS

File System	**DBMS**
Collection of data that is stored in the file/directory	Software that allows you to create, define, maintain, and control access to a DB
Data integrity is not easier to maintain here	Data integrity is easier to maintain in DBMS
It doesn't handle data redundancy	Handles data redundancy
Data is inconsistent as redundancy is present	Data is consistent as there is no redundancy

File System VS RDBMS:

Field --- coulmn/attribute.
File ----- table, entity, relation
Record --- Row/tuple

2. Relation between two files is maintained programmatically
2. Relation between two or more tables is maintained by keys

3. More Programming is required in case of file system.
3. Less Programming is required, (we write different queries to get the data)

student : id (Primary key), name, prn(foreign Key), emailid
result : PRN(primary key), makrs, rank

select student.id, student.name, student.prn, result.marks, result.rank from student, result
where student.prn=result.prn
and student.name= 'RITIK'

4. More time required for the s/w development.
4. less time required for the s/w development.

5. High Network traffic.;
5. Low Network Traffic

6.Slow and Expansive (we need to different programms/scripts)
6.Faster and less expansive

select * from student limit 100;

7.Processing of the data is takes place at client machine in case of file systems
7.In case RDBMS the processing of the data is takes place at the server.

8. File level Locking is there in case of file syetem,
8. In case of RDBMS, row level locking there,

9. Not suitable for multiple users
9. Suitable for Multiple users

10. Security concern : OS level security
10. Multiple level of secuirty :
username and password for mysql
command level secuirty: we grant the permission to perticular user; (to insert , to select, to create).

Types of DBMS: relational, object-relational, NoSQL

RDBMS: MySQL

Object-Relational DBMS:
It does everything a relational database can do, but it allows developers to store complex data types—like maps, audio, or custom programming objects—directly inside a table cell, complete with inheritance and custom functions.

E.g., Postgres
Real-World Scenario: A Real Estate App

NoSQL: NoSQL" stands for *Not Only SQL*. It drops the rigid rules, rows, and columns to handle massive amounts of unstructured, fast-changing data across hundreds of servers.

MongoDB: an example
E,g Instagram

DBMS vs Database

Database	**DBMS**
1.It is a collection of interrelated data that can be structured or unstructured.	1.DBMS is a software that allows users to create, define, maintain and control access to a database.
2. These represent physical or electronic locations where data is stored.	2.It serves as a software interface that interacts between user and database.
3. Ex. Customer relation management(CRM), Enterprise Resource Planning(ERP), Accounting System etc.	3. Ex.MySQL, MongoDB, Oracle
4. Benefits we get in databases are -data management, data protection, data sharing.	4. Benefits we get are: data sharing, data integration, data security, data performance
5. Users in a database are: software developer, user administrator, data analyst, Business users, end user	5. Users in DBMS are: software developer, user administrator, data analyst, Business users

Data Models

1. Conceptual : It provides the high level view of the entities.

2. Logical: A logical data model organises these into a detailed database struture

3. Physical: Specifes how that structure will actually be implemented and stored in particular DBMS.

e.g library management: Books, Members, Author, Borrow, Publisher, book_type

Logical: Books : id, name, author,

Members: MemberId, Name, Role,

Author: Aut_Id, Auth_name, Auth_address

Borrow: ID, Date,

Publisher: ID, Date of Publisher

Book_type:History, science,

MySQL is structured query language which is further is divided into 4 parts-

1. **DQL (Data Query Language)** is a subset of SQL used to **retrieve data from a database**.

   DQL is used for **fetching or querying data**, not modifying it.

   The primary command is:

   SELECT → used to get data from one or more tables

1. **Data Manipulation Language(DML)-**

- DML statements affect the record in the table and perform operations like selecting, insertion, deleting, updating and modifying any existing record.
- The commands used here are insert, update and delete.

2. Data Definition Language(DDL)-

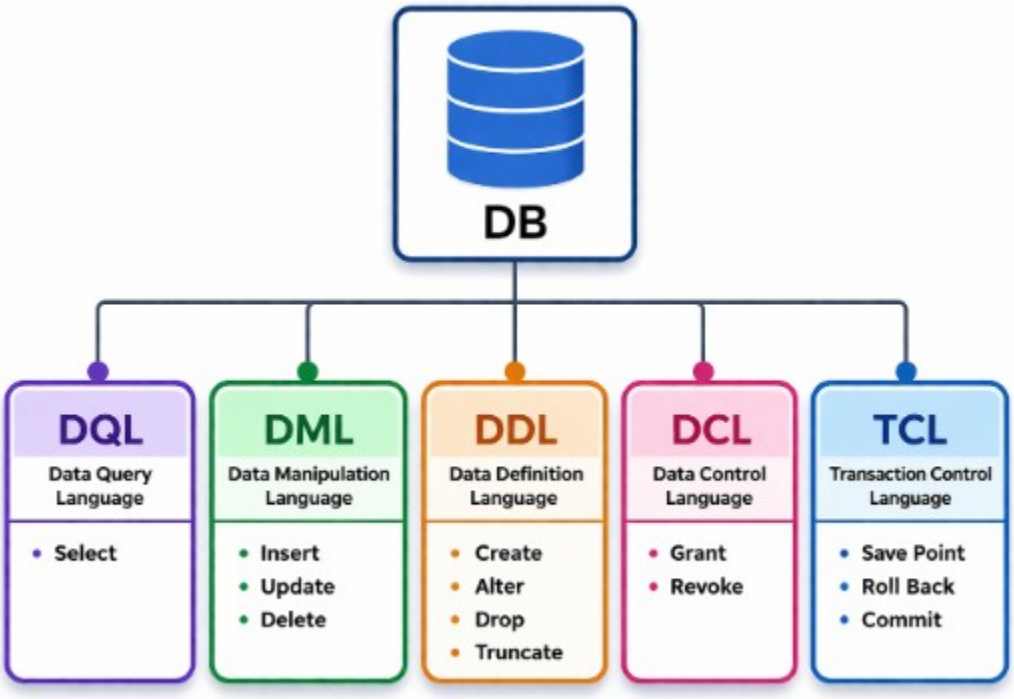
- This is used to alter, modify database or table structure and schema. These statements are used to handle storage of database objects or design.
- The commands used here are create, alter, truncate and drop

3. Data control Language(DCL)-

- DCL controls the level of access that users have on database objects.
- Commands like grant and revoke are used in DCL.

4. Transaction Control Language(TCL)-

- These allow you to control and manage transactions to maintain the integrity of data within the system.
- Commands like save point, commit and rollback are used in TCL.



```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| aug25              |
| aug26              |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| test1              |
+--------------------+
7 rows in set (0.00 sec)

mysql> use aug26;
Database changed
mysql> show tables;
+------------------+
| Tables_in_aug26  |
+------------------+
| books_management |
| student          |
| student2         |
+------------------+
3 rows in set (0.00 sec)

mysql> drop table student;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> create table student ( id int, name varchar(15), profile json);
Query OK, 0 rows affected (0.02 sec)

mysql> desc student;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| id      | int         | YES  |     | NULL    |       |
| name    | varchar(15) | YES  |     | NULL    |       |
| profile | json        | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)

mysql> insert into student ( id, name, profile ) values (1, 'Yash', '{"skills": ["Python", "SQL"], "score": 92, "address": {"city": "Mumbai"} , "DOB
": "11-12-1996"}');
Query OK, 1 row affected (0.01 sec)

mysql> select * from student;
+------+------+-----------------------------------------------------------------------------------------------------------+
| id   | name | profile                                                                                                   |
+------+------+-----------------------------------------------------------------------------------------------------------+
|    1 | Yash | {"DOB": "11-12-1996", "score": 92, "skills": ["Python", "SQL"], "address": {"city": "Mumbai"}} |
+------+------+-----------------------------------------------------------------------------------------------------------+
1 row in set (0.00 sec)

mysql> select name, profile->>'$.score' as score from student;
+------+-------+
| name | score |
+------+-------+
| Yash | 92    |
```

```
mysql> create table student ( id Int, Name char(30), age INT, course char(10), City varchar(10));
Query OK, 0 rows affected (0.02 sec)

mysql> desc student;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| id      | int         | YES  |     | NULL    |       |
| Name    | char(30)    | YES  |     | NULL    |       |
| age     | int         | YES  |     | NULL    |       |
| course  | char(10)    | YES  |     | NULL    |       |
| City    | varchar(10) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

```
mysql> insert into student (id, Name, age, course, city) values ( 2, 'Ritesh', 25, 'Mysql', 'Satar
Query OK, 1 row affected (0.00 sec)

mysql> select * from student;
+------+--------+------+--------+--------+
| id   | Name   | age  | course | City   |
+------+--------+------+--------+--------+
|    1 | ANIKET |   30 | JAVA   | Mumbai |
|    2 | Ritesh |   25 | Mysql  | Satara |
+------+--------+------+--------+--------+
2 rows in set (0.00 sec)

mysql> insert into student (id, age, Name, course, city) values ( 2, 25, 'Gauri', 'DotNet', 'Pune'
Query OK, 1 row affected (0.00 sec)

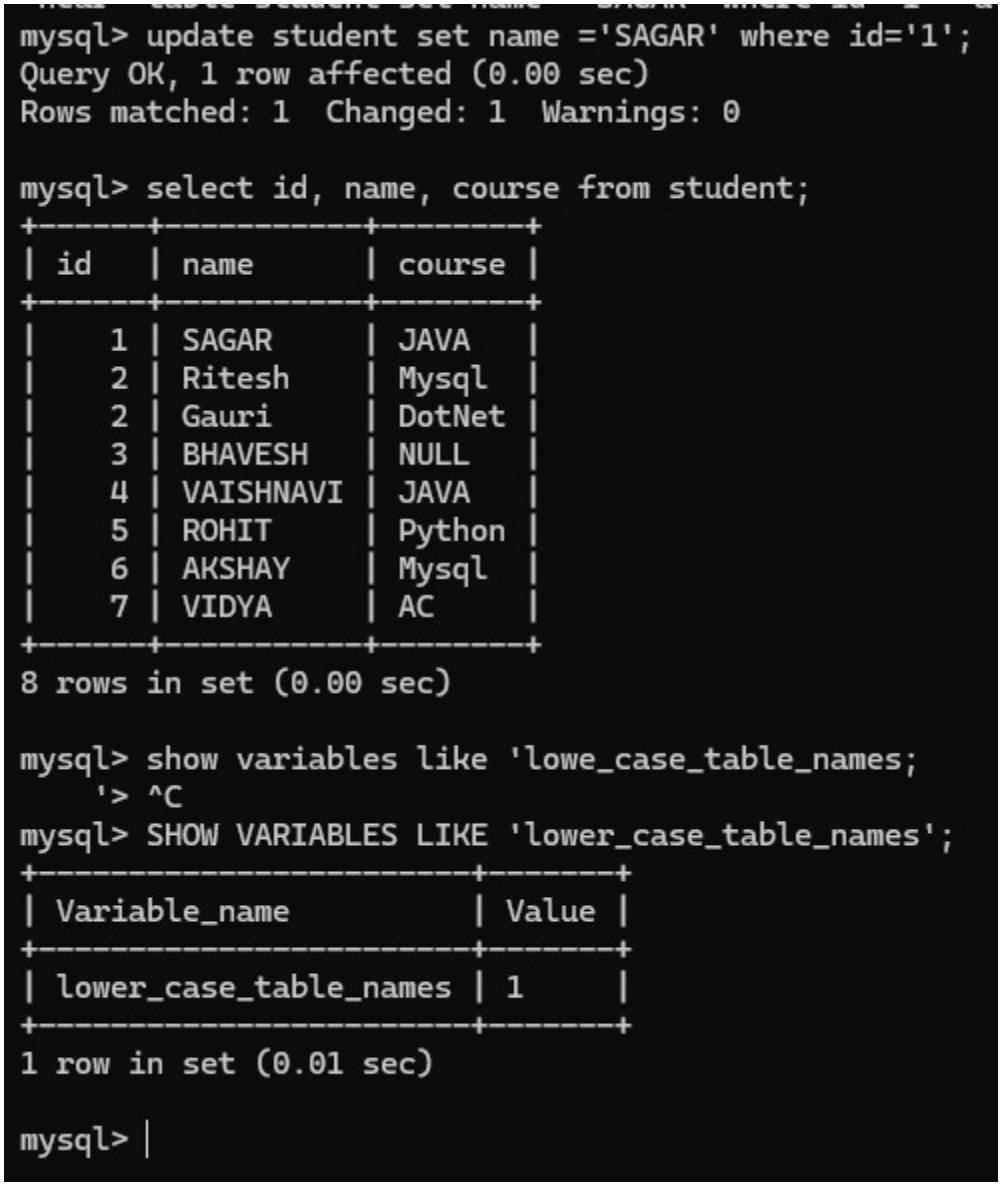
mysql> select * from student;
+------+--------+------+--------+--------+
| id   | Name   | age  | course | City   |
+------+--------+------+--------+--------+
|    1 | ANIKET |   30 | JAVA   | Mumbai |
|    2 | Ritesh |   25 | Mysql  | Satara |
|    2 | Gauri  |   25 | DotNet | Pune   |
+------+--------+------+--------+--------+
3 rows in set (0.00 sec)

mysql> insert into student (id, name, city) values ( 3, 'BHAVESH', 'SATARA');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into student (id, Name, age, course, city) values ( 4, 'VAISHNAVI', 23, 'JAVA', 'Satara'),
    -> (5, 'ROHIT', 28, 'Python', 'Kolhapur'), (6, 'AKSHAY', 30, 'Mysql', 'Sangli'), (7, 'VIDYA' , 25, 'AC', 'PUNE');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> select * from student;
+------+-----------+------+--------+----------+
| id   | Name      | age  | course | City     |
+------+-----------+------+--------+----------+
|    1 | ANIKET    |   30 | JAVA   | Mumbai   |
|    2 | Ritesh    |   25 | Mysql  | Satara   |
|    2 | Gauri     |   25 | DotNet | Pune     |
|    3 | BHAVESH   | NULL | NULL   | SATARA   |
|    4 | VAISHNAVI |   23 | JAVA   | Satara   |
|    5 | ROHIT     |   28 | Python | Kolhapur |
|    6 | AKSHAY    |   30 | Mysql  | Sangli   |
|    7 | VIDYA     |   25 | AC     | PUNE     |
+------+-----------+------+--------+----------+
8 rows in set (0.00 sec)

mysql> select id, name, course from student;
+------+-----------+--------+
| id   | name      | course |
+------+-----------+--------+
|    1 | ANIKET    | JAVA   |
|    2 | Ritesh    | Mysql  |
|    2 | Gauri     | DotNet |
|    3 | BHAVESH   | NULL   |
|    4 | VAISHNAVI | JAVA   |
|    5 | ROHIT     | Python |
|    6 | AKSHAY    | Mysql  |
|    7 | VIDYA     | AC     |
+------+-----------+--------+
8 rows in set (0.00 sec)
```



```
mysql> update student set name ='SAGAR' where id='1';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select id, name, course from student;
+------+----------+--------+
| id   | name     | course |
+------+----------+--------+
|    1 | SAGAR    | JAVA   |
|    2 | Ritesh   | Mysql  |
|    2 | Gauri    | DotNet |
|    3 | BHAVESH  | NULL   |
|    4 | VAISHNAVI| JAVA   |
|    5 | ROHIT    | Python |
|    6 | AKSHAY   | Mysql  |
|    7 | VIDYA    | AC     |
+------+----------+--------+
8 rows in set (0.00 sec)

mysql> show variables like 'lowe_case_table_names;
    '> ^C
mysql> SHOW VARIABLES LIKE 'lower_case_table_names';
+------------------------+-------+
| Variable_name          | Value |
+------------------------+-------+
| lower_case_table_names | 1     |
+------------------------+-------+
1 row in set (0.01 sec)

mysql>
```

SHOW VARIABLES LIKE 'lower_case_table_names';